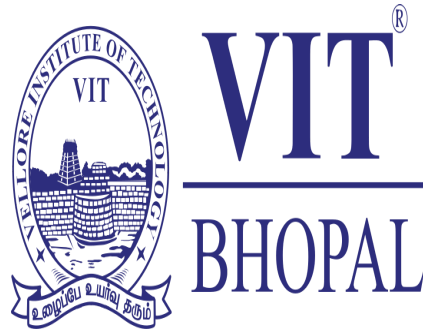


VIT BHOPAL UNIVERSITY



Student Result Management System

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1. Introduction

Student Result Management System is a simple Python based project used to store and manage student marks and results. The project is console based and uses basic Python concepts such as variables, functions, lists, dictionaries, loops and conditional statements.

2. Problem Statement

Managing student marks manually can take time and may lead to calculation mistakes. Calculating total marks, percentage and grades for multiple students can also become difficult. This project provides a simple way to enter student marks, calculate results and display useful result information.

3. Objectives

- Store student details and marks.
- Calculate total marks and percentage.
- Calculate student grade.
- Search for a student.
- Find the highest percentage.
- Calculate the class average.

4. Functional Requirements

1. Student Management: Add, display and search student records.
2. Result Processing: Calculate total marks, percentage and grade.
3. Result Analysis: Find the highest percentage and class average.

5. Non-Functional Requirements

- Usability: Simple menu and clear output.
- Reliability: Correct calculation of totals, percentages and grades.
- Error Handling: Handles invalid choices and empty records.
- Maintainability: Functions keep tasks separate and easy to modify.

6. System Architecture

USER → MAIN MENU / PROGRAM → STUDENT MANAGEMENT / RESULT PROCESSING / RESULT ANALYSIS
→ STUDENT RECORDS

7. Workflow / Process

START → Display Menu → Enter Choice → Perform Selected Operation → Display Result → Return to Menu → Exit

8. Use Case

The user can add students, display students, search for a student, find the highest percentage, calculate class average and exit the system.

9. Sequence

User → Select operation → System asks for input → System processes data → System displays result.

10. Component Design

Main Menu; Student Management; Result Processing; Result Analysis; Student Records.

11. Design Decisions

Basic Python concepts were selected to demonstrate fundamental programming skills. Functions divide the program into smaller tasks. A dictionary stores student records and lists store marks and result information. A menu-driven approach makes the system easy to use.

12. Implementation

The project is implemented in Python. A while loop keeps the menu running. Functions are used for adding, displaying and searching students, finding the highest percentage and calculating class average. If-elif-else statements are used for grade calculation and for loops process records.

13. Sample Results

Example: Rahul — Python: 80, Maths: 75, EVS: 85, Total: 240, Percentage: 80.0, Grade: B. The program can also search for Rahul, find the highest percentage and calculate the class average.

14. Testing

Test cases include adding a student, displaying records, searching for an existing student, searching for an unknown student, finding the highest percentage, calculating class average, invalid menu choice and exiting.

15. Challenges Faced

Understanding how to store multiple student records; calculating percentage correctly; finding the highest percentage; creating a menu using a while loop; dividing the program into functions.

16. Learnings

Learned functions, lists, dictionaries, loops, conditions, arithmetic operations, menu-driven programming, basic testing and project documentation.

17. Future Enhancements

Add student roll numbers; add more subjects; store records in a file; add attendance; generate individual report cards; add a graphical interface.

18. Conclusion

The Student Result Management System provides a simple way to manage student results using basic Python programming concepts. It demonstrates functions, lists, dictionaries, loops, conditions and arithmetic operations in a practical project.

19. References

1. Python documentation.
2. CSE course notes.
3. VITyarthi Build Your Own Project guidelines